

The future of ^{our} jobs: Will AI take our place

the impact of technology on children

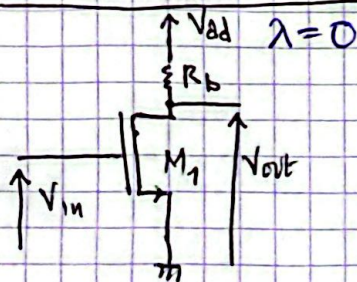
Work-study programs

Technology and Communication

$$I_2 = \frac{W_2/L_2}{W_1/L_1} \cdot I_1$$

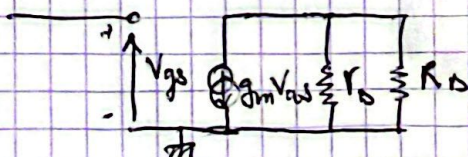
VMWare Workstation pro.

password: ameziane



$$C_{ir} = \frac{V_{out}}{V_{in}} = \frac{-g_m V_{gs} R_b}{V_{gs}} = -g_m R_b$$

$\lambda \neq 0$

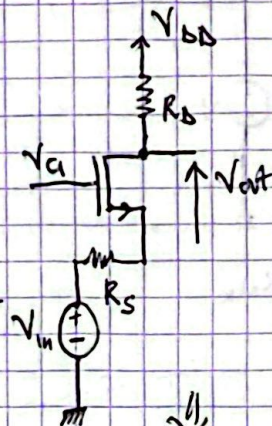


$$C_i = -(r_o // R_b) g_m \quad V_{out} = -(r_o // R_b) g_m V_{gs} \quad V_{in} = V_{gs}$$

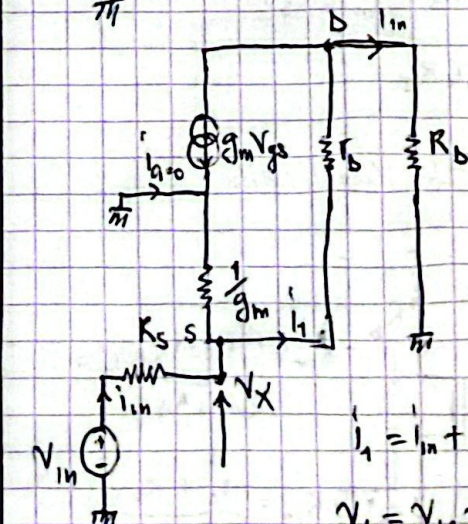
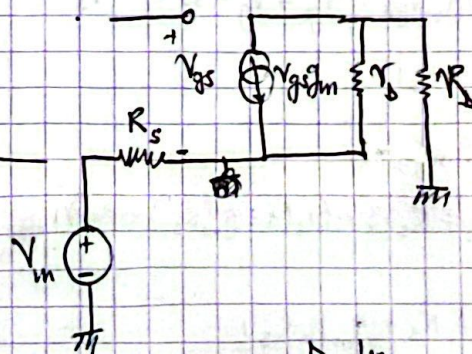
$$V_{DD} = 20V$$

$$I_D = \frac{1}{2}$$

* BASIC cell / Common Gate



$\lambda \neq 0$

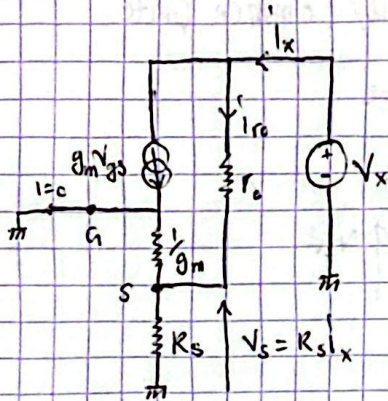


$$\begin{aligned} i_1 &= i_{in} + g_m V_{gs} \\ V_x &= V_{in} + R_s i_{in} \\ \text{avec } V_{gs} &= -V_x, \text{ alors } V_x = r_o i_1 + R_b i_{in} \\ V_x &= r_o (i_{in} - g_m V_x) + R_b i_{in} \\ V_x (1 + r_o g_m) &= r_o i_{in} + R_b i_{in} \\ V_x &= \frac{r_o + R_b}{1 + r_o g_m} i_{in} \Rightarrow R_{in} = \frac{V_x}{i_{in}} = \frac{r_o + R_b}{1 + g_m r_o} \end{aligned}$$

* si $r_o \gg R_D =$

$$R_{in} = \frac{1}{g_m}$$

* Output Resistance: R_{out}



$$R_{out} = \frac{V_x}{I_x}$$

$$I_{r_o} = I_x - g_m V_{gs} / V_{gs} = V_g - V_S = -V_S$$

$$\text{with } V_S = R_S I_x$$

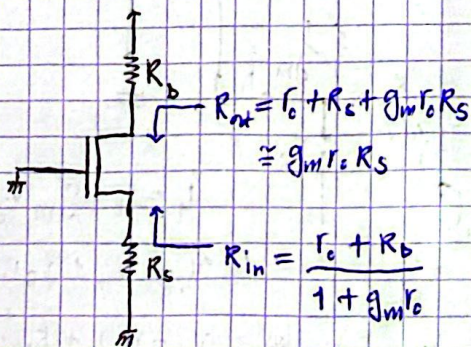
$$I_{r_o} = I_x + g_m R_S I_x$$

$$V_x = r_o I_{r_o} + R_S I_x = (r_o (1 + g_m R_S) + R_S) I_x$$

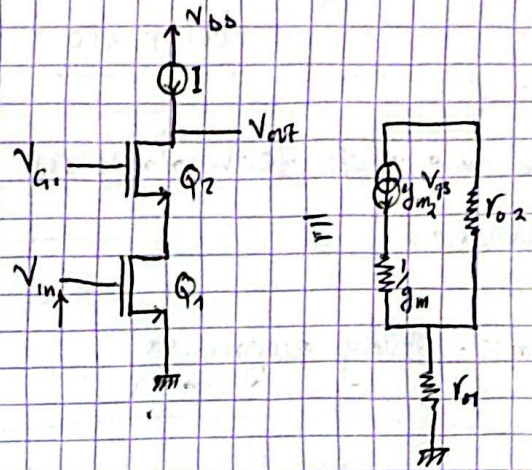
$$R_{out} = r_o + R_S + g_m r_o R_S$$

$$R_{out} \approx g_m r_o R_S$$

* Common Gate Modelisation



* Increasing the Gain



$$\frac{1}{g_{m2}} // r_{o2} + r_{o1} = \frac{r_{o2} \cdot 1}{\frac{g_{m2}}{r_{o2}} + 1} + r_{o1}$$

$$\frac{r_{o2}}{1 + g_{m2} r_{o2}} + r_{o1} = \frac{r_{o2} + r_{o1} (1 + g_{m2} r_{o2})}{1 + g_{m2} r_{o2}}$$

On a augmenté la résistance